1 Search Models

1.1 Exercise 7.1

A McCall Model with a chance of unemployment. The Bellman equation is as follows

$$v(w) = \max_{a\in\{\text{Accept, Reject}\}} \{\frac{w}{1-\beta}, \beta(\phi(c+\bar{V}) + (1-\phi)\mathbb{E}\left[v(w')\right])\} \tag{1.1}$$

where $\bar{V} = \frac{(1+\beta\phi)c+\beta(1-\phi)\mathbb{E}[v(w')]}{1-\beta\phi}$ denotes the continuation value of being unemployed.[1] $v(w)$ denotes the value function conditional on receiving an offer. $\mathbb{E}\left[v(w')\right] = \int_0^B v(w')dF(w')$ denotes the expected value conditional on receiving an offer.

1.2 Exercise 7.2

A McCall Model with two offers per period. Since $F(x) = \Pr(z \le x)$, the cumulative distribution function of the maximum of two random variables i.i.d. drawn from $F(x)$ is $G(x_m) = \Pr(x < x_m)\cdot \Pr(x < x_m) = [F(x_m)]^2$.

$$v(w_m) = \max_{a\in\{\text{Accept, Reject}\}} \left(\frac{w_m}{1-\beta}, c + \beta\mathbb{E}_{\max}[v(w'_m)]\right) \tag{1.2}$$

where $\mathbb{E}_{\max}[v(w'_m)] = \int_0^B v(w'_m)\, d[F(w'_m)]^2$. Since $\mathbb{E}_{\max}[v(w'_m)] = 2\int_0^B v(w'_m)f(w'_m)F(w'_m)dw' > \int_0^B v(w'_m)f(w'_m)dw'$,[2] we have

$$\bar{w}_{\max} = (1-\beta)\big(c + \beta\mathbb{E}_{\max}[v(w'_m)]\big) > (1-\beta)\big(c+\beta\mathbb{E}[v(w'_m)]\big) = \bar{w}.$$

■

1.3 Exercise 7.3

A McCall Model with n iid offers per period. Generalize the formulation in Exercise 7.2, we have

$$v(w) = \max_{a\in\{\text{Accept, Reject}\}} \left(\frac{w}{1-\beta}, c + \beta\mathbb{E}_{\max}[v(w')]\right) \tag{1.3}$$

[1] I obtain $\bar{V}$ by solving the following Bellman equation: $\bar{V} = c + \beta\Big(\phi(c+\bar{V}) + (1-\phi)\mathbb{E}\left[v(w')\right]\Big)$

[2] We apply the monotonicity of $v(x)$ and $2F(x)-1$ to finish the proof:

$$\begin{aligned}
&2\int_0^B v(w'_m)f(w'_m)F(w'_m)dw' - \int_0^B v(w'_m)f(w'_m)dw' \\
=&\int_0^B v(w'_m)(2F(w'_m)-1)dF(w') \\
=&\mathbb{E}\left[v(w'_m)(2F(w'_m)-1)\right] \\
=&\underbrace{\mathrm{Cov}\left[v(w'_m), 2F(w'_m)-1\right]}_{>0} + \underbrace{\mathbb{E}[v(w'_m)]\mathbb{E}[2F(w'_m)-1]}_{>0}
\end{aligned}$$

where $\mathbb{E}_{\max}[v(w')] = n\int_0^B v(w')F^{n-1}(w')f(w')dw'$.

1.4 Exercise 7.4

A McCall Model with a random number of offers per period. Denote the probability of n in this period but m in the next period as π_{mn}.

Part a. The Bellman Equation is given as follows.

$$v(w,n) = \max_{\{\text{Accept, Reject}\}} \left(\frac{w}{1-\beta}, c + \beta(\mathbb{E}[v(w',n')]) \right) \tag{1.4}$$

where w is the best offer. $\mathbb{E}[v(w',n')] = \sum_{m=1}^N \pi_{mn} \int_0^B mF(w')^{m-1}v(w',n'=m)dw'$ with $\sum_m \pi_{mn} = 1$.

Part b. We characterize the reservation wage $\bar{w}(n)$ as follows.

$$\bar{w}(n) = (1-\beta)\beta\left[\sum_{m=1}^N \pi_{mn} \int_0^B v(w',n'=m)dF(w')\right] + c(1-\beta) \tag{1.5}$$

where $\sum_{m=1}^N \pi_{mn} \int_0^B v(w',n'=m)dF(w') \equiv \mathbb{E}[v(w',m)]$ is the marginal expected value function given the number of offers arrived.

Note that $\sum_m \pi_{mn}\mathbb{E}[v(w',m)]$ is a function of n, it follows that $\bar{w}(n)$ is also a function of n.

1.5 Exercise 7.5

A McCall Model with an endogenous number of offers per period. In this exercise we consider a case where the worker endogenously choose the number of offers $n \in 1,2,\ldots,N$ before the wages are drawn. Formally, the Bellman Equation is given by:

$$v(w_m) = \max_{a\in\{\text{Accept, Reject}\}} \left(\frac{w_m}{1-\beta}, c + \beta \max_{n\in\{1,\ldots,N\}} \left[-k(n) + \int_0^B v(w'_m)d[F(w'_m)]^n \right] \right) \tag{1.6}$$

1.6 Exercise 7.7

A McCall Model with variable labor supply. We first show that the optimal labor supply is a time invariant policy function, which solely depends on w. Let $\tilde{v}(w)$ be the value function of accepting an offer at wage w. Since the state variable w does not change across periods, the Bellman equation that characterize the optimal labor supply is given by:

$$\tilde{v}(w) = \max_{l\in[0,1]} [u(w(1-l),l) + \beta\tilde{v}(w)] \tag{1.7}$$

Denote the optimal labor supply given wage w by $l^*(w)$, we have

$$u(w(1-l^*), l^*) = (1-\beta)\tilde{v}^*(w)$$

Therefore, the optimal labor supply $l^*(w)$ does not depend on time (because $\tilde{v}^*(w)$ is time-invariant.)[3]

Finally, we can specify the Bellman equation using the optimal labor supply policy $l^*(w)$ as follows.

$$v(w) = \max_{\{\text{Accept, Reject}\}} \left\{ \frac{u\Big(w(1-l^*(w)), l^*(w)\Big)}{1-\beta}, u(c,1) + \beta \int_0^B v(w')dF(w') \right\} \tag{1.8}$$

Note that the preservation wage structure in the Bellman equation above is given by following equation.

$$u\Big(w(1-l^*(w)), l^*(w)\Big) = (1-\beta)(u(c,1) + \beta\mathbb{E}[v(w')]) \tag{1.9}$$

1.7 Exercise 7.8

Wage growth rate and the reservation wage The Bellman Equation is given by

$$v(w) = \max_{\{\text{Accept, Reject}\}} \left\{ \frac{w}{1-\beta\phi}, c + \beta \int_0^B v(w')dF(w') \right\} \tag{1.10}$$

Denote the reservation wage by $\bar{w}$, we have

$$\begin{aligned} \frac{\bar{w}}{1-\beta\phi} &= c + \beta\Bigg[\int_0^{\bar{w}} \frac{\bar{w}}{1-\beta\phi} dF(w') + \int_{\bar{w}}^B \frac{w'}{1-\beta\phi} dF(w')\Bigg] \\ &= c + \beta\Bigg[\frac{\bar{w}}{1-\beta\phi} \int_0^B dF(w') + \int_{\bar{w}}^B \frac{w'-\bar{w}}{1-\beta\phi} dF(w')\Bigg] \end{aligned} \tag{1.11}$$

$\Rightarrow$

$$\underbrace{\frac{(1-\beta)\bar{w}}{1-\beta\phi} - c}_{g(\bar{w})} = \underbrace{\beta \int_{\bar{w}}^B \frac{w'-\bar{w}}{1-\beta\phi} dF(w')}_{h(\bar{w})} \tag{1.12}$$

We can use the graph below to visually show that $\frac{\partial \bar{w}}{\partial \phi} < 0$.

[3] Here I assume that the worker will stick to one optimal labor supply level under multiple equilibria.

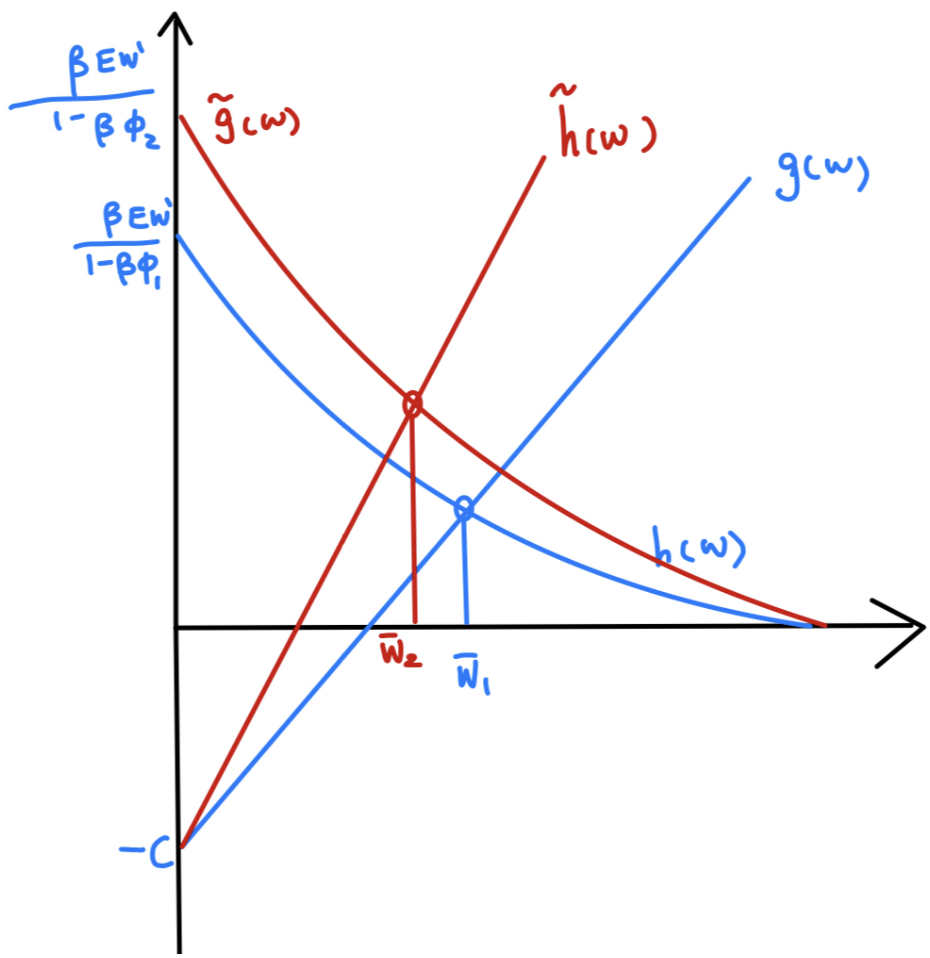

Figure 1.1: Wage growth and reservation wage

Here, I finish the proof in a more rigorous way which requires additional condtion for β.

Define a function $G(\bar{w}) = g(\bar{w}) - h(\bar{w}) = \frac{(1-\beta)\bar{w}}{1-\beta\phi} - c - \frac{\beta}{1-\beta\phi}\int_{\bar{w}}^{B}(w' - \bar{w})dF(w')$. It follows that $G'(\bar{w}) = \frac{1-\beta F(\bar{w})}{1-\beta\phi} > 0$. Suppose $\phi_2 = \kappa\phi_1$ with $\kappa > 1$, and $G(\bar{w}_1; \phi = \phi_1) = 0$. In order to show $\bar{w}_2 < \bar{w}_1$, we need to show that $G(\bar{w}_1; \phi = \kappa\phi_1) > 0$ because $G(w)$ is increasing.

However, the condition for $G(\bar{w}_1; \phi = \kappa\phi_1) > 0$ is that $\bar{w}_1$ must be sufficiently high. The following derivation will show that if $\bar{w}_1 = B$, then $\bar{w}_2 < \bar{w}_1$ without additional condition. Contrarily, if $\bar{w}_1 \to 0$, we will require $\beta \to 0$ to ensure $\bar{w}_2 < \bar{w}_1$. In particular, the condition is given by $\beta < \frac{\bar{w}_1}{\bar{w}_1 F(\bar{w}_1) + \mathbb{E}_{w' > \bar{w}_1}[w']}$.[4]

$$
\begin{aligned}
G(\bar{w}_1; \phi = \phi_1) - G(\bar{w}_1; \phi = \kappa\phi_1) &= \left[\bar{w}_1(1-\beta) - \beta\int_{\bar{w}_1}^{B}(w' - \bar{w}_1)dF(w')\right]\underbrace{\left(\frac{1-\beta}{1-\beta\phi} - \frac{1-\beta}{1-\beta\kappa\phi}\right)}_{\Delta\phi<0} \\
&= \Delta\phi\left[\bar{w}_1(1-\beta) - \beta\int_{\bar{w}_1}^{B}(w' - \bar{w}_1)dF(w')\right] \\
&= \Delta\phi\left[\bar{w}_1(1-\beta) - \beta\int_{\bar{w}_1}^{B}w'dF(w') + \beta\int_{\bar{w}_1}^{B}\bar{w}_1 dF(w')\right] \\
&= \Delta\phi\left[\bar{w}_1(1-\beta) - \beta\int_{\bar{w}_1}^{B}w'dF(w') + \beta\bar{w}_1(1 - F(\bar{w}_1))\right] \\
&= \Delta\phi\left[\bar{w}_1 - \beta\underbrace{\int_{\bar{w}_1}^{B}w'dF(w')}_{\equiv\mathbb{E}_{w'>\bar{w}_1}[w']} - \beta\bar{w}_1 F(\bar{w}_1)\right]
\end{aligned}
\tag{1.13}
$$

1.8 Exercise 7.10

Finite horizon and mean-preserving spread. First, we consider a situation where there is no unemployment compensation.

[4] When $\bar{w}_1 = B$, the condition is automatically satisfied because $\frac{\bar{w}_1}{\bar{w}_1 F(\bar{w}_1) + \mathbb{E}_{w'>\bar{w}_1}[w']} = 1$. When $\bar{w}_1 \to 0$, we have $\frac{\bar{w}_1}{\bar{w}_1 F(\bar{w}_1) + \mathbb{E}_{w'>\bar{w}_1}[w']} \to 0$.

2 Mean-preserving Spread

2.1 Answer to Question 2.1

(a) According to the definition of mean-perserving spread (MSP) in **?**, Y is a MSP from X if $Y = X + Z$ where $\mathbb{E}[Z|X] = 0$. Therefore,

$$
\begin{aligned}
\mathrm{Var}(Y) &= \mathrm{Var}(X+Z) \\
&= \mathrm{Var}(X+Z) \\
&= \mathrm{Var}(X) + \mathrm{Var}(Z) + 2\mathrm{Cov}(X,Z) \\
&= \mathrm{Var}(X) + \mathrm{Var}(Z) + 2\Big(\mathbb{E}(XZ) - \mathbb{E}(X)\mathbb{E}(Z)\Big) \\
&= \mathrm{Var}(X) + \mathrm{Var}(Z) + 2\Big(\mathbb{E}(X\mathbb{E}(Z|X)) - \mathbb{E}(X)\mathbb{E}(\mathbb{E}[Z|X])\Big) \\
&= \mathrm{Var}(X) + \mathrm{Var}(Z) \\
&\geq \mathrm{Var}(X)
\end{aligned}
$$

■

(b) Y can be generated by X without using MSP while satisfies the properties of $\mathbb{E}(X) = \mathbb{E}(Y)$ and $\mathrm{Var}(X) = \mathrm{Var}(Y)$ as follows.

$$
\begin{bmatrix} X \\ Y \end{bmatrix} \sim N\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} \right) \tag{2.1}
$$

Remark: The process above is not a MSP because $\mathbb{E}[Z|X] \neq 0$.

Transitivity of mean-preserving spreads.

(a) Suppose $X2$ is contructed from $X1$ by MSP, and $X3$ is contructed from $X2$ by MSP. Then we have

$$
\begin{aligned}
X2 &= X1 + Z1 \\
X3 &= X2 + Z2
\end{aligned}
$$

where $\mathbb{E}[Z1|X1] = 0$, and $\mathbb{E}[Z2|X2] = 0$. Therefore,

$$
X3 = X1 + Z1 + Z2
$$

Now if follows that

$$\begin{aligned}
\mathbb{E}[(Z1+Z2)|X1] &= \mathbb{E}[(Z1+Z2)|X1] \\
&= \mathbb{E}[Z2|X1] \\
&= \mathbb{E}[Z2|X2, Z1] \\
&= \mathbb{E}[Z2|X2] (\text{because } Z1 \text{ is independent of } Z2) \\
&= 0
\end{aligned}$$

■

(b) Denote $(F, G) \in S$ if distribution functions F and G satisfy single-crossing property. The figure below shows that $(F1, F2) \in S$ and $(F2, F3) \in S$, but $(F1, F3) \notin S$.

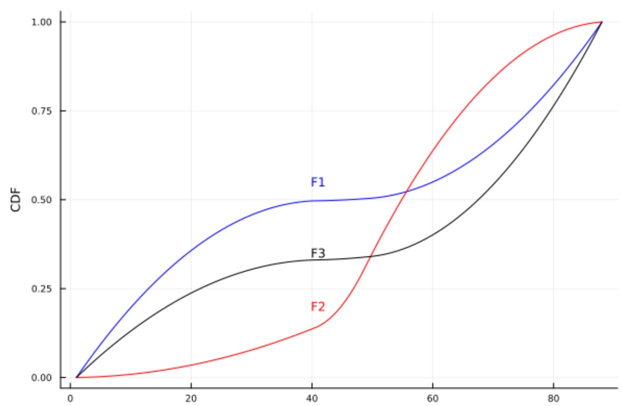

Figure 2.1: Single-crossing property is not transitive

3 Numerical implementation of the finite-horizon search problem

Answer to Question 3.1

The Bellman equation is given by

$$v_t(w) = \max_{\{\text{Accept, Reject}\}} \left\{ \frac{w(1-\beta^{T-t+1})}{1-\beta}, c + \beta \int_0^B v_{t+1}(w') dF(w') \right\}, \quad \forall t = 0, 1, \ldots, T \tag{3.1}$$
$$v_{T+1}(w) = 0$$

Answer to Question 3.2

Since $v_{T+1}(w) = 0$, the maximization problem is given by

$$v_T(w) = \max_{\{\text{Accept, Reject}\}} \{w, c\}$$

Therefore,

$$v_T(w) = \max\{w, c\}$$

Answer to Question 3.3

We use induction to prove that the optimal policy is a reservation wage. When $t = T$, $\bar{w}_T = c$. When $t = T - 1$, the Bellman Equation is:

$$v_{T-1}(w) = \max_{\{\text{Accept, Reject}\}} \{w + \beta w, c + \beta(F(c)c + \int_c^B w' dF(w'))\}$$

Therefore, $\bar{w}_{T-1} = \frac{c+\beta(F(c)c+\int_c^B w' dF(w'))}{1+\beta}$, which is a constant.

Answer to Question 3.4

Inspired by the answers above, we can characterize the indifference condition as follows.

$$\frac{\bar{w}_t(1-\beta^{T-t+1})}{1-\beta} = c + \beta\left(F(\bar{w}_{t+1})c + \int_{\bar{w}_{t+1}}^B w' dF(w')\right)$$

Answer to Question 3.5

Yes, this problem has well-defined solution even if $\beta > 1$ as we can solve it using backward induction.

Answer to Question 3.6

See my iPad.

Answer to Question 3.7

See my iPad.

We now solve the model numerically in two different ways. In what follows, use the following benchmark parameter values: $T = 50$, $\beta = 0.97$, $B = 1$, $c = 0.3$. For the distribution $F(w)$ of wage offers, start with $w \sim \text{Beta}(\alpha, \beta)$, with parameters $\alpha = 2, \beta = 2$. Recall that the Beta (α, β) distribution has the density

$$f(w) = \frac{w^{\alpha-1}(1-w)^{\beta-1}}{\mathrm{B}(\alpha, \beta)},$$

where $\mathrm{B}(\alpha, \beta)$ is the Beta function.

Answer to Question 3.8

My code is available at:

https://github.com/BillJunbiaoChen/GA1025_MacroTheory/tree/main/ProblemSets/PS1_computation

The expected value function at $t+1$ is given by

$$E[v_{t+1}(w')] = \sum_{\{i: w_i < \bar{w}_{t+1}\}} \bar{w}_{t+1} f_i + \sum_{\{i: w_i \geq \bar{w}_{t+1}\}} w_i f_i$$

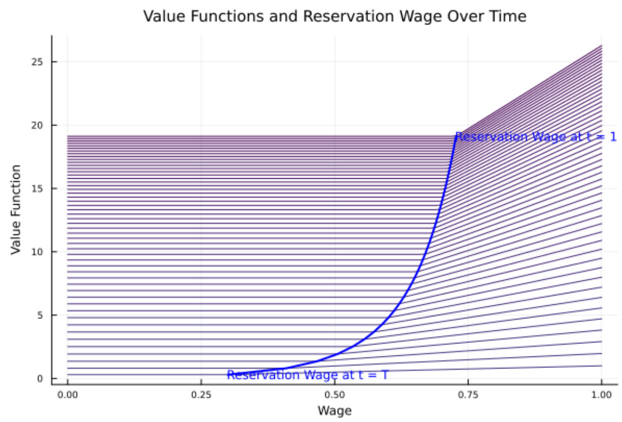

Figure 3.1: Value functions and reservation wages

Answer to Question 3.12

Recall that the indifference condition for the reservation wage in the infinite horizon is given by

$$\begin{aligned}
\frac{\bar{w}}{1-\beta} &= c + \beta \left[\int_0^{\bar{w}} \frac{\bar{w}}{1-\beta} dF(w') + \int_{\bar{w}}^{B} \frac{w'}{1-\beta} dF(w') \right] \\
&= c + \beta \left[\frac{\bar{w}}{1-\beta} + \int_{\bar{w}}^{B} \frac{w' - \bar{w}}{1-\beta} dF(w') \right]
\end{aligned}$$

$\Rightarrow$

$$\bar{w} - c = \frac{\beta}{1-\beta} \int_{\bar{w}}^{B} (w' - \bar{w}) dF(w')$$

Define a function $g(w) = w - c - \frac{\beta}{1-\beta} \int_w^B (w' - w) dF(w')$, and then apply discretization listed above, we have

$$\tilde{g}(w) = w - c - \frac{\beta}{1-\beta} \sum_{\{i: w_i > w\}} (w_i - w) f_i \tag{3.2}$$

To solve the equation above, I apply `Roots` package in `Julia`. The table compares the reservation wages (i.e., the root in equation 3.2) obtained via different algorithms and box constraints.

Interval	Algorithm	Time (s)	Root
$[0, 1]$	Secant	0.020007	0.74870
$[0, 1]$	Bisection	0.029884	0.74870
$[0, 1]$	A42	0.027841	0.74870
$[0.3, 1]$	Secant	0.006241	0.74870
$[0.3, 1]$	Bisection	0.008908	0.74870
$[0.3, 1]$	A42	0.009308	0.74870

Table 3.1: Reservation wage in infinite horizon computed by root-finding algorithms